

Nafisa Islam Rifa

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 Nafisa Islam Rifa |  NafisaIslamRifa |  Nafisa_Rifa221 |  Google Scholar
London, United Kingdom

EDUCATION

- **University Of Greenwich** September 2025 - October 2026 (Expected)
MSc in Data Science London, United Kingdom
 - Current Average: 71% (Distinction), 120/180 credits completed
- **United International University** February 2022 - Aug 2025
Bachelor of Computer Science and Engineering Dhaka, Bangladesh
 - GPA: 3.95/4.00
- **Rajshahi College** May 2019
Higher Secondary Certificate Rajshahi, Bangladesh
 - GPA: 5.00/5.00
- **Govt PN Girls' High School** March 2017
Secondary School Certificate Rajshahi, Bangladesh
 - GPA: 5.00/5.00

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [C.1] (Accepted) Iftexhar Mahmud, Puja Rani Das, **Nafisa Islam Rifa**, Irfan Hossain, Raiyan Rahman and Md. Farid Dewan, "Multimodal Emotion Recognition Using Visual and Thermal Image Fusion: A Deep Learning Approach", IEEE ICCIT 2024 [IEEE Conference].
- Worked on multimodal fusion of visual and thermal images
 - Performed dataset preprocessing and model experimentation
 - Evaluated deep learning model performance

RESEARCH EXPERIENCE

- **Vision-Language Models for Automated Radiology Report Generation** 2025-2026
Supervisor: Dr. Peter Soar, Senior Lecturer in Computer Science, University Of Greenwich, London, United Kingdom
 - Developed a **fully autonomous visual grounding system** using cross-attention to generate real-time anatomical heatmaps for clinical interpretability.
 - Optimized training with **clinical masking and weighted sampling** to mitigate data imbalance, achieving a **0.89 BERTScore** for report accuracy.

PROJECTS

- **Build Large Language Model from Scratch** 2025 
 - Developed a GPT-like Large Language Model from scratch based on the book *Build a Large Language Model (From Scratch)* by Sebastian Raschka.
- **Build Multimodal Vision-Language Model from Scratch** 2025 
 - Implemented a multimodal Vision-Language Model based on **PaliGemma**, integrating a **SigLIP Vision Transformer encoder** with **GemmaForCausalLM** using a custom **PaliGemmaForConditionalGeneration** architecture and projector for visual-text embedding alignment.
- **Future Electricity Demand Prediction in Bangladesh** 2025 
 - Collected 10 years of electricity demand data from the PGCB website and built a structured dataset including fuel usage, temperature, and other key features.
 - Developed hybrid deep learning models (CNN-LSTM and RNN-LSTM) and benchmarked them against XGBoost, achieving improved forecasting accuracy and stronger model reliability.
- **Network Intrusion Detection** 2025 
 - Built a network intrusion detection system using the augmented UNSW-NB15 dataset with feature engineering for robust performance.
 - Compared Logistic Regression, Decision Tree, Random Forest, SVM, and XGBoost models, with Random Forest achieving the highest accuracy.

SKILLS

- **Programming Languages:** C, C++, Python, R, Java, SQL, PHP ,JavaScript
- **Data Science & Machine Learning:** NumPy, Pandas, Matplotlib, PyTorch, TensorFlow.
- **Databases:** MySQL, MongoDB,Cassandra

ACHIEVEMENTS

SSC Scholarship (Secondary School Certificate, Bangladesh) 2017

- Recipient of the SSC Scholarship in Bangladesh for exceptional performance in the Secondary School Certificate (SSC) examination, with an overall score of 87.93% and a board rank of 191 among 152,783 candidates.

HSC Scholarship (Higher Secondary Certificate, Bangladesh) 2019

- Recipient of the HSC Scholarship in Bangladesh for exceptional performance in the Higher Secondary Certificate (HSC) examination, with an overall score of 91% and a board rank of 473 among 123,616 candidates.

Academic Scholarship, United International University (UIU)

- Awarded a 100% merit-based tuition fee waiver for 5 out of 10 trimesters at United International University (UIU) for ranking within the top 2% of the program. Maintained a perfect GPA of 4.00/4.00 across all awarded trimesters.

Best Technical Presentation Award, ICCIT 2024 2024

- Awarded Best Technical Presentation Award at the International Conference on Computer and Information Technology (ICCIT) 2024 for outstanding presentation and research contribution.

LANGUAGE PROFICIENCY

- **English:** Proficient, IELTS Academic – Overall Score: 7.5 (Listening: 8.0, Reading: 7.5, Writing: 6.5, Speaking: 7.5)
- **Bengali:** Native Language

REFERENCES

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